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The Effect of Aromatherapy Massage on Sleep Quality

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Outline:

I. Introduction

1. Problem: Poor sleep quality in both physical and mental health.
2. Theoretical background:
    - The Importance and Causes of Sleep Quality.
    - Aromatherapy massage is a complementary therapy that uses essential oils during massage to encourage relaxation and comfort.
3. Research question: What is the effect of aromatherapy massage on sleep quality?
4. Definitions of Variables: sleep quality, aromatherapy massage.

II. Methodology

1. Keywords that guided the research: Aromatherapy massage, sleep quality.
2. Years covered: 2016–2026.
3. Sources used: Scopus.
4. Inclusion and exclusion.

III. Results

1. Grouping of Articles:
    - *Two groups:* one group including studies conducted in Japan, and one group including studies conducted in Turkey.

2. Studies in Japan: (Kawabata et al., 2020).

    a) Purpose & Questions/Hypothesis.

    b) Sample

    c) Instruments and procedures

    d) Results

    e) Practical implications

3. Studies in Turkey: (Özlü & Bilican, 2017; Ayik & Özden, 2018; Akgun et al., 2021).

    a) Purpose & Questions/Hypothesis.

    b) Sample

    c) Instruments and procedures

    d) Results

e) Practical implications

IV. Discussion

1. An Overview of the findings in relation to the research question:

   Three out of four studies showed a positive effect of aromatherapy massage on sleep quality.

2. Differences:
   - Difference in sample size.
   - Difference in sample age.
   - Difference in sample location.
   - Difference in instruments measuring sleep quality.

3. Interpretations of the findings of the studies:
   - Variance in outcomes across studies.
   - Limited generalizability
     - Limited locations.
     - Limited sample size.

4. Recommendations for future research:
   - Increasing sample size.
   - Conducting studies in different countries.

References

I. Introduction:

Sleep quality plays an essential role in both physical and mental health, and poor sleep has been associated with a range of adverse outcomes, including depression, anxiety, cardiovascular disease, metabolic disorders, and impaired cognitive functioning (Baranwal et al., 2023). Various everyday factors can negatively influence sleep quality, such as psychological stress, excessive use of electronic devices before bedtime, unhealthy lifestyle behaviors, and sleep disorders including insomnia and sleep apnea (Baranwal et al., 2023; Xu et al., 2024). Given the widespread impact of sleep disturbances and the limitations of conventional treatments, there has been growing interest in complementary and non-pharmacological approaches that may support sleep quality (Chang et al., 2017). Aromatherapy massage is one such approach, involving the application of essential oils through massage, allowing for both dermal absorption and inhalation, and is commonly associated with relaxation and comfort (Tang et al., 2021). Despite increasing attention to aromatherapy massage in sleep-related research, the relationship between aromatherapy massage and sleep quality remains insufficiently explored, particularly in terms of establishing a clear and consistent understanding of its role as a complementary intervention, highlighting the need for further investigation (Xu et al., 2024).

The aim of this review is to determine the impact of aromatherapy massage on sleep quality, and the research question is: What effect does aromatherapy massage have on sleep quality?

In this review, aromatherapy massage (the independent variable) is defined in the first study as using blended essential oils like lavender, orange, or a combination of both (Kawabata et al., 2020). In other study aromatherapy massage is defined as lavender oil ratio (1:1) was applied (Ozlu & Bilican, 2017). In another study, it's defined as a back massage that is applied using 5% lavender oil diluted in sweet almond oil by a certified researcher in back massages (Ayik & Özden, 2018). In the last study, the aromatherapy massage was defined as back massages using a mixture of lavender and almond oil, employing techniques like effleurage and petrissage (Akgun et al., 2021). Sleep quality (the dependent variable) in the first study was defined using five components: awakening, sleep depth, sleep latency, efficiency, and perceived sleep quality (Kawabata et al., 2020). In other study, sleep quality was defined by the depth of night sleep, sleep latency time, waking frequency, awake time, wake-up time, the effect of noise, and levels of ambience on the quality of sleep (Ozlu & Bilican, 2017). In another study, sleep quality is defined as consisting of 5 elements: perceived sleep depth, sleep latency (time to fall asleep), number of awakenings, efficiency (percentage of time awake) and sleep quality (Ayik & Özden, 2018).In the last study the Sleep quality was defined by seven components including sleep latency, duration, disturbances, sleep efficiency, subjective sleep quality, use of sleep medication, and daytime dysfunction (Akgun et al., 2021).

II. Methodology:

The Scopus database was used to find the research studies included in this review. The keywords that were used were aromatherapy and sleep quality. Some delimiters were used, including research studies written in English, and the years were limited from 2016 to 2026. To ensure relevance to specific research settings, which gave the final number of 10 results.

Out of the 10 research studies, two were excluded because they were secondary. Four studies were primary; however, they were excluded because they were irrelevant to the research question. One study evaluated the integrated yoga program in patients with inflammatory bowel disease rather than aromatherapy massage and sleep quality. Another study discussed the effectiveness of an alternative nursing strategy on sleep patterns in coronary intensive care during hospitalization instead of aromatherapy massage and sleep quality. In another study, it was testing aromatherapy massage and its effect on quality of life rather than quality of sleep. The last study was trying complementary interventions on intensive care patients to see if it impacted the quality of sleep instead of using aromatherapy massage. In addition, one was primarily relevant, but it was excluded because there were too many variables.

The remaining three studies were relevant, so they were included in this literature review (Kawabata et al., 2020; Ayik & Ozden, 2018; Ozlu & Bilican, 2017). One study conducted was a supplementary study obtained from Google Scholar (Akgun et al., 2021).

III. Results:

All four articles included in this literature review can be grouped into two groups according to the results of the articles. The first group reports that aromatherapy massage has a positive effect on sleep quality. This group includes three studies (Ozlu & Bilican, 2017; Ayik & Ozden, 2018; Akgun et al., 2021). The second group reports that aromatherapy massage does not have an effect on sleep quality. This group includes (Kawabata et al., 2020).

The first study belonging to the first group and conducted by Ozlu & Bilican (2017) aimed at comparing two groups to determine the effect of aromatherapy massage on sleep quality and psychological parameters among post-surgical patients admitted to the intensive care unit. The research question addressed what effect aromatherapy massage has on sleep quality and psychological parameters in post-surgical intensive care patients. The sample consisted of 60 patients aged 18 years or older, including both males and females, who were recruited from the intensive care unit of Ataturk University Research Hospital in Turkey. The selection criterion used was convenient sampling. Regarding the instruments, sleep quality was measured using the Richards-Campbell Sleep Questionnaire (RCSQ), which evaluates depth of night sleep, sleep latency, waking frequency, awake time, wake-up time, the effect of noise, and ambient conditions. The questionnaire consists of six items scored from 0 to 100, where scores between 0–25 indicate poor sleep quality and scores between 76–100 indicate excellent sleep quality. Regarding the procedure, baseline questionnaires and interviews were conducted for both groups. Lavender oil diluted at a 1:1 ratio was applied to the experimental group at 22:00 for 10–15 minutes using 3–5 mL of oil, while the control group received routine nursing care only. Measurements were taken before the intervention and at 15, 30, 60, and 120 minutes following application. The study duration was three months, from November 2013 to January 2014, and the type of procedure was a pretest–posttest experimental design with a control group. The results showed a statistically significant improvement in overall sleep quality in the experimental group compared to the control group, as reflected by higher total RCSQ scores following the intervention ($p < 0.001$). Improvements were observed across multiple sleep components, including depth of night sleep, reduced sleep latency, decreased frequency of awakenings, and better perceived sleep quality. In contrast, no significant improvement in sleep quality was observed in the control group over the same time periods. However, the generalizability of the results is limited due to the use of convenience sampling and the inclusion of patients from a single hospital. As for practical implications, the researchers suggested incorporating aromatherapy massage before bedtime as part of routine nursing care to improve sleep quality in intensive care patients.

The second study belonging to the first group and conducted by Ayik & Özden (2018) aimed at comparing two groups to determine the effect of aromatherapy massage with lavender oil on sleep quality among patients undergoing colorectal surgery. The study hypothesized that aromatherapy massage would positively affect patients' sleep quality during the preoperative period. The sample consisted of 80 patients aged 18 years or older, including both males and females, who were admitted to the General Surgery Unit of a University Hospital in Turkey. The selection criteria were not explicitly stated. Regarding the instruments, sleep quality was measured using the Richards-Campbell Sleep Questionnaire (RCSQ), which assesses sleep depth, sleep latency, number of awakenings, sleep efficiency, and overall sleep quality. Scores range from 0 to 100, with scores between 0–25 indicating very poor sleep quality and scores between 76–100 indicating very good sleep quality. Regarding the procedure, participants were divided into an experimental group and a control group. The experimental group was informed about the purpose of the study between 16:00 and 17:00 hours and completed the RCSQ. A certified researcher then applied a 10-minute back massage using 5% lavender oil between 19:00 and 21:00 hours on the evening before surgery. The control group received routine care only. All participants completed the RCSQ again on the morning of surgery. The study duration was between January 25th and May 31st, and it covered the preoperative evening until the morning of surgery, and the type of procedure was a pretest–posttest experimental design with a control group. The results revealed a statistically significant difference in RCSQ scores between the experimental and control groups, with the experimental group showing higher sleep quality scores ($p < 0.05$), supporting the study hypothesis. However, limitations related to the experimental design and sampling methods restricted the generalizability of the findings. As for practical implications, the researchers recommended the use of aromatherapy massage during the preoperative period to enhance sleep quality and patient comfort, and they emphasized the importance of training nurses to incorporate aromatherapy massage within holistic patient care.

The third study belonging to the first group and conducted by Akgun et al., (2021) aimed at determining the impact of aromatherapy massage on sleep quality of older people living in nursing homes to treat their sleep problem. The researchers hypothesized that aromatherapy massage would improve sleep quality and reduce sleepiness. The sample consisted of 15 older people above 65 years old from the nursing home in Istanbul, Turkey. As for the participants' gender, there were 8 females and 7 males. The selection criterion was not mentioned. Regarding the instruments, the researchers measured subjective sleep quality, sleep latency, duration of sleep activity, sleep disorder, and use of opiate daytime function disorder with Pittsburgh Sleep Quality Index (PSQI). Scores range from 0 to 21, and a total score above 5 indicates poor sleep quality. Regarding the procedure, First, the researchers met with 62 older people in the nursing home, but only 15 agreed to participate in the study. At the start, all participants were evaluated using the PSQI, and two groups of seven were formed based on their scores. For two weeks, each participant received a back massage with lavender and almond oil for 20 minutes, three times a week, making a total of six sessions. Before they received 53 hours of training from 8 different branches from "Homemade Aromatherapy", and All stages of the process were carefully followed including room setup, equipment preparation, and both practitioner and individuals. Ethical approval and written permission were obtained from Üsküdar University, and the participants were fully informed about the purpose of the study before providing consent. The study duration was between May 21 and June 21, 2018. The type of procedure was pre-post design. The results showed a strong overall improvement in sleep quality after the aromatherapy massage intervention. The total PSQI score was statistically significant ($p = 0.001$). Several PSQI subcomponents, including sleep latency, sleep duration, and habitual sleep efficiency, also improved significantly ($p < 0.05$). These results support the hypothesis that aromatherapy massage improves sleep quality in older adults. However, the generalizability of the findings is limited because only 15 volunteers from a single nursing home were included. As for practical implications, the researchers suggest that gentle and calming non-drug approaches, such as aromatherapy massage, should be applied to help improve sleep quality for older people.

After reviewing studies showing that aromatherapy massage has a positive effect on sleep quality, studies reporting no effect of aromatherapy massage on sleep quality will also be tackled.

The first study belonging to the second group and conducted by Kawabata et al., (2020) aimed to examine the effectiveness of aromatherapy massage on the quality of sleep and its help on body fatigue determining the impact of aromatherapy massage on sleep quality of older people living in nursing homes to treat their sleep problem. The researchers' question was to see the effect of aromatherapy massage on sleep quality. The sample consisted of 57 people 18 years or older from cancer patients from the National Hospital Organization Kyoto Medical Center in Kyoto, Japan. As for the participants' mixed gender. The selection criteria were not mentioned. Regarding the instruments, sleep quality was measured using the Richards-Campbell Sleep Questionnaire (RCSQ), which assesses sleep depth, sleep latency, number of awakenings, sleep efficiency, and overall sleep quality. Scores range from 0 to 100, with scores between 0–25 indicating very poor sleep quality and scores between 76–100 indicating very good sleep quality. Regarding the procedure, First, the researcher registered the patients' after being eligible, then they randomly assigned patients to the control group or experiment group based on sex using a computer to divide them. The patients were tested on the day of the aromatherapy massage and the next day, the aromatherapy massage was only performed on patients who stayed longer than 3 days in the hospital, the massage was performed between 8 and 9 pm. Each session lasted 30 minutes and the patients could choose either lavender oil, orange oil or a mix of both, they were medical grade oils, and the therapist had been certified from the EPA & AEAJ, they performed effleurage massage on patients from the palm to the elbow region and from the sole of foot to knee area. The study duration was from January 2018 to March 2019. The type of procedure was pre-post design. The result of the study found that a single 30-minute session of nighttime aromatherapy massage did not significantly improve sleep quality in palliative care patients compared to the control group, as measured by the Richards-Campbell Sleep Questionnaire (RCSQ). Analysis of covariance showed no statistically significant difference in total RCSQ scores ($P = 0.60$) or in any of its subscales (e.g., sleep depth $P=0.51$, sleep latency $P = 0.61$). Post hoc analysis suggested that older patients and those with poorer performance status showed moderate improvement in sleep quality after aromatherapy massage ($r=0.45$ and $r=0.40$, respectively), but the overall effect was not significant. As for practical implications, not mentioned.

IV. Discussion:

In answering the research question, it can be concluded that aromatherapy massage generally has different results, where there were three studies that had a positive effect of aromatherapy massage on sleep quality (Özlü & Bilican, 2017; Ayik & Özden, 2018; Akgun et al., 2021), and one study showed that aromatherapy massage had no effect on sleep quality (Kawabata et al., 2020). In one study, hospitalized patients who received aromatherapy massage showed a significant improvement in sleep quality compared to the control group (Özlü & Bilican, 2017). In another study, patients undergoing colorectal surgery who received lavender aromatherapy massage experienced significantly better sleep quality than the control group (Ayik & Özden, 2018). In another study, older adults living in nursing homes showed improved sleep quality from the repeated sessions of aromatherapy massage compared to the control group (Akgun et al., 2021). In another study, no significant improvement in sleep quality was observed among palliative care patients after a single session of aromatherapy massage (Kawabata et al., 2020).

Among the 4 studies, some differences were found. Three out of four studies had a sample size of 60 or less persons (Kawabata et al., 2020; Özlü & Bilican, 2017; Akgun et al., 2021), the other study had a sample size greater than 60 (Ayik & Özden, 2018). Another difference is the location of the study, three out of four studies were conducted in Turkey (Özlü & Bilican, 2017; Ayik & Özden, 2018; Akgun et al., 2021), the last of the four was conducted in Japan (Kawabata et al., 2020). The ages of the persons in the experiment was also a point of difference, two studies had an age range of 60 and above (Ayik & Özden, 2018; Akgun et al., 2021), while the other 2 studies had an age range of 18 years or older (Kawabata et al., 2020; Özlü & Bilican, 2017). The last point of difference found was the instrument used to evaluate the results, three out of four studies used RCSQ (Kawabata et al., 2020; Özlü & Bilican, 2017; Ayik & Özden, 2018), while the last of the four used PSQI (Akgun et al., 2021).

It can be interpreted that the results differ across the four studies. Three studies reported that aromatherapy massage had a positive effect on sleep quality (Özlü & Bilican, 2017; Ayik & Özden, 2018; Akgun et al., 2021). In contrast, one study found that aromatherapy massage had no effect on sleep quality (Kawabata et al., 2020). These differences in results may be attributed to variations in study context, as the studies reporting positive effects were conducted in Turkey, whereas the study reporting no effect was conducted in Japan.

It is recommended that future studies use a larger sample size (above 100 participants), as all the reviewed studies included fewer than 100 participants. Increasing the sample size would enhance the generalizability of the findings. Additionally, future research should be conducted in a wider range of countries rather than being limited to Turkey and Japan. Since three of the four studies were conducted in Turkey and only one in Japan, conducting studies across different countries would improve the generalizability and accuracy of the results.

References

Akgun, M., OZ, M., & Uymaz, P. (2021). The Effect of Aromatherapy Massage on Older People's Sleep Quality and Sleepiness. *ASEAN Journal of Psychiatry, 22*(10), 1–14.

Ayik, C., & Özden, D. (2017). The effects of preoperative aromatherapy massage on anxiety and sleep quality of colorectal surgery patients: A randomized controlled study. *Complementary Therapies in Medicine, 36*, 93–99. https://doi.org/10.1016/j.ctim.2017.12.002

Baranwal, N., Yu, P. K., & Siegel, N. S. (2023). Sleep physiology, pathophysiology, and sleep hygiene. *Progress in Cardiovascular Diseases, 77*, 59–69. https://doi.org/10.1016/j.pcad.2023.02.005

Chang, Y.-Y., Lin, C.-L., & Chang, L.-Y. (2017). The Effects of Aromatherapy Massage on Sleep Quality of Nurses on Monthly Rotating Night Shifts. *Evidence-based Complementary and Alternative Medicine*, *2017*, 1-8. https://doi.org/10.1155/2017/3861273

Kawabata, N., Hata, A., & Aoki, T. (2020). Effect of Aromatherapy Massage on Quality of Sleep in The Palliative Care Ward: A Randomized Controlled Trial. *Journal of Pain and Symptom Management, 59*(6), 1165–1171. https://doi.org/10.1016/j.jpainsymman.2020.01.003

Özlü, Z. K., & Bilican, P. (2017). Effects of aromatherapy massage on the sleep quality and physiological parameters of patients in a surgical intensive care unit. *African Journal of Traditional, Complementary and Alternative Medicines, 14*(3), 83–88. https://doi.org/10.21010/ajtcam.v14i3.9

Tang, Y., Gong, M., Qin, X., Su, H., Wang, Z., & Dong, H. (2021). The Therapeutic Effect of Aromatherapy on Insomnia: A meta-analysis. *Journal of Affective Disorders, 288*, 1–9. https://doi.org/10.1016/j.jad.2021.03.066

Xu, K., Wang, S., Ji, Q., Ni, Y., & Liu, T. (2024). Effects of aromatherapy on sleep quality in older adults: A meta-analysis. *Medicine*, *103*(49), 1-7. https://doi.org/10.1097/MD.0000000000040688